N016AV (S/N: W14031869222) — Cellular Modem Hardware Failure Report

Immfly Ground Operations / IT Session: September 15, 2026

Field	Value
Machine	**N016AV** (S/N: W14031869222)
Affected component	Sierra Wireless EM7565 Cellular Modem (Internal USB Port `1-4.1` / M.2 PCIe Socket)
Initial symptom	Cellular data path loss; system initially authenticated via PAP, but suffered sudden interface termination: `Failed to resolve interface "ppp0": No such device` followed by `Failed to open /dev/modemPPP: No such file or directory`. Subsequent commands confirmed complete disappearance from USB bus.
USB identification	**ABSENT** — Completely missing from USB topology (no `1199:9091` nor `1199:9090` detected by kernel). Internal hub port `1-4.1` is entirely dead.
Root cause	Terminal hardware electrical failure, internal power management IC (PMIC) collapse, or M.2 electrical disconnect on the Sierra Wireless module. The device fails to negotiate electrical handshakes across PCI bus rescan (`echo 1 > /sys/bus/pci/rescan`) and full host cold reboots. `rfkill` reports no software or hardware blocks.
Comparison	Shares identical root failure mechanics with **N109AV (S/N: W03021891423)**. Distinct from recovery/bootloader lockups (N136AV / N060AV in QDL mode `1199:9090`) or storage block degradation (N067AV, N078AV, N084AV). In N016AV, peripheral devices on the same physical controller (FTDI bridges and Teensyduino) operate normally, isolating the breakdown directly to the cellular modem module.
Recommended action	**Physical Sierra Wireless EM7565 modem replacement required.** Line maintenance work order needed.

NARRATIVE SUMMARY

Server unit **N016AV (S/N: W14031869222)** exhibited fatal failure of cellular connectivity. Diagnostic executio with `modem_fix.py` initially reported an outdated firmware release (`SWI9X50C_01.08.04.00`) operating i `QMI` mode. While the module negotiated initial registration on the Movistar network and performed PA authentication, the data link immediately collapsed with unrecoverable device loss: `Failed to resolv interface "ppp0": No such device` and `Failed to open /dev/modemPPP`.

Detailed kernel topology inspection via `lsusb -t` and `lsusb | grep -i 1199` revealed that the Sierr Wireless module (Vendor ID `1199`) vanished completely from internal USB hub `1-4` (port `1-4.1`) Concurrently, adjacent peripheral devices on the same physical hub controller—specifically FTDI dual seria bridges (Port 2, `Dev 5`) and Teensyduino CDC ACM controllers (Port 3, `Dev 6`)—remained fully enumerate and operational. This unequivocally localizes the fault to the modem's internal circuitry or M.2 socket.

Extensive low-level recovery procedures were performed, including force-triggering kernel PCI bus re enumeration (`echo 1 > /sys/bus/pci/rescan`) and completing a host machine cold reboot to power cycle internal buses. Following reboot, the modem remained entirely undetected at the electrical level, an `rfkill list` confirmed that no software or hardware RF kill switches were active. The module is electricall unresponsive and permanently out of service.

N016AV (S/N: W14031869222) — Diagnostic Timeline & Console Appendix

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DIAGNOSTIC FINDINGS & SEQUENCE OF EVENTS

1. **Diagnostic Interrogation & Initial State:**
   Automated tool `modem_fix.py` detected the modem in QMI mode with outdated firmware (`SWI9X50C_01.08.04.00`) and confirmed network registration on Movistar.
2. **Abrupt Bus Disconnect During Link Staging:**
   During PPP daemon handshake and IPCP exchange, the physical serial bridge dropped completely: `Failed to open /dev/modemPPP: No such file or directory`, causing immediate failure of `resolvectl` with `Failed to resolve interface "ppp0": No such device`.
3. **USB Bus Topology Audit:**
   Inspection of `lsusb -t` verified that while Port 2 (FTDI) and Port 3 (Teensy CDC ACM) of internal Hub `1-4` remained functional, Port `1-4.1` (designated for the Sierra Wireless module) reported zero attached devices.
4. **Electrical Bus Rescan Failure:**
   Executing a manual PCI bus rescan (`echo 1 > /sys/bus/pci/rescan`) produced no kernel enumeration events. `lsusb | grep -i 1199` returned null.
5. **Cold System Reboot Verification:**
   A complete system power cycle via `reboot` was executed to reinitialize motherboard voltage regulators and PCIe/M.2 power rails. Post-reboot checks confirmed the modem remained fully absent from both `lsusb` and `dmesg`.
6. **RF Kill Exemption:**
   `rfkill list` confirmed no active hardware or software RF blocks across existing wireless subsystems, confirming the root cause is electrical or physical hardware failure.

CONCLUSION AND RECOMMENDATION

The Sierra Wireless EM7565 modem in aircraft server **N016AV (S/N: W14031869222)** has experienced tota electrical/hardware failure. It does not enumerate on the USB bus nor respond to bus rescans or cold systen reboots. Software-level remediation has been entirely exhausted.

Recommended action: Issue an urgent Line Maintenance Work Order to **physically replace the Sierra Wireless EM7565 cellular modem** in aircraft server unit **N016AV (S/N: W14031869222)**.

APPENDIX: CONSOLE DIAGNOSTIC OUTPUT & BUS TOPOLOGY

```
[AVA:N016AV] USB bus topology inspection confirming dead modem port:
# lsusb -t
/:  Bus 02.Port 1: Dev 1, Class=root_hub, Driver=xhci_hcd/4p, 5000M
/:  Bus 01.Port 1: Dev 1, Class=root_hub, Driver=xhci_hcd/4p, 480M
    |__ Port 2: Dev 2, If 0, Class=Vendor Specific Class, Driver=ftdi_sio, 12M
    |__ Port 2: Dev 2, If 1, Class=Vendor Specific Class, Driver=ftdi_sio, 12M
    |__ Port 3: Dev 3, If 0, Class=Wireless, Driver=btusb, 12M
    |__ Port 3: Dev 3, If 1, Class=Wireless, Driver=btusb, 12M
    |__ Port 4: Dev 4, If 0, Class=Hub, Driver=hub/4p, 480M
        |__ Port 3: Dev 6, If 1, Class=CDC Data, Driver=cdc_acm, 480M
        |__ Port 3: Dev 6, If 0, Class=Communications, Driver=cdc_acm, 480M
        |__ Port 2: Dev 5, If 0, Class=Vendor Specific Class, Driver=ftdi_sio, 12M

[AVA:N016AV] Device lookup, dmesg, and rfkill status post-reboot:
```

```
# lsusb | grep -i 1199
(no output)

# dmesg | grep -i -E "1199|sierra|disconnect" | tail -n 15 && rfkill list
0: hci0: Bluetooth
        Soft blocked: no
        Hard blocked: no
1: phy0: Wireless LAN
        Soft blocked: no
        Hard blocked: no

# echo 1 > /sys/bus/pci/rescan && lsusb | grep -i 1199
(no output)
```